

Tutorial 0 – Week 1

We'll be starting at the 10 minute mark

Agenda

1. TA Introductions
2. Logistics
3. Install Anaconda Navigator
4. Set up a Folder Structure for APS106
5. VSCode/Jupyter Notebook
6. UofT JupyterHub
7. Questions?

Introduction – TA



Mehdi Salakhi

(TUT105, TUT107)

Current position: Postdoctoral Fellow (Defended my PhD at UofT in Nov 2025)

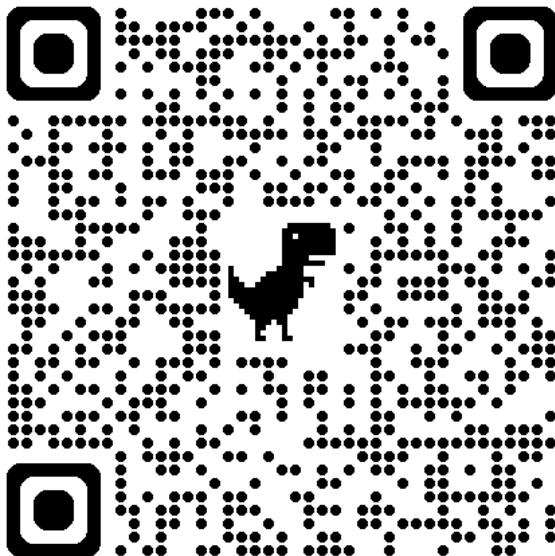
Research/other interests: Clean Energy, Multiphysics Modeling, Football

Tutorial Logistics

- Tutorials are for your benefit – **no grading**
 - We will review previous weeks labs & lecture content
 - From Week 2, we will focus on **coding problems!**
- Be sure to **ask lots of questions** and have Python open.
We are here to help you!
- Questions outside of tutorial time?
 - Post to **Piazza** – all TAs/instructors and your peers can answer questions quickly

Online Tutorials/Office Hours Survey

- URL - <https://forms.gle/s8VTXuMrVezatbQU7>



Why code?¹

- ***"Coding is the closest thing we have to a superpower!"*** – Drew Houston, Dropbox Co-founder
 - Coding empowers you to solve real-world problems creatively.
 - Code is the driving force behind technological innovations.
 - Coding opens doors to a plethora of career opportunities.
 - Every line of code is a step towards building solutions that make a difference.



- Images Created by DALL-E, an AI Image Generation Model developed by OpenAI

Why code?²

- Real-World Examples
 - Ever wondered how Instagram suggests reels you might like?
 - Coding and algorithms enable early disease detection, enhance diagnostic accuracy, and save lives.
 - Virtual Reality (VR) and Augmented Reality (AR) experiences are crafted through intricate code, with gaming, simulation, and immersive experiences depending on the skills of programmers.

AI-generated (DALL-E)



Source



Source



Why code?³

- Space agencies, such as NASA, heavily depend on coding for space exploration missions (spacecraft navigation, data analysis, etc.).
- Autonomous vehicles use image recognition for safe navigation, as cameras capture real-time surroundings and algorithms identify objects, pedestrians, and other vehicles.
- CAD and computational software tools: SolidWorks, Ansys, COMSOL, ...
- Regardless of background, coding is always an achievable skill
- **NEVER THINK THAT YOU “CAN’T UNDERSTAND” CODING**



Source



Source

"Programming isn't about what you know; it's about what you can figure out." - Chris Pine, Learn to Program



What would you want to ideally learn how to do with computers?

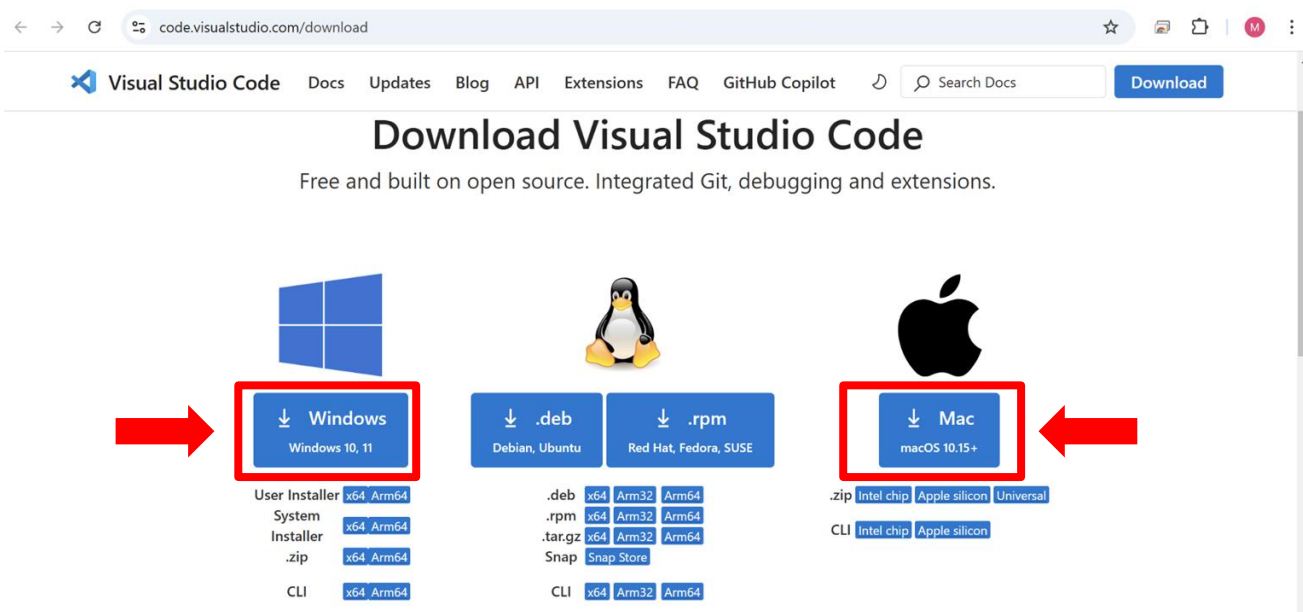


Coding experience?

Install VS Code

Install VS Code¹

- You can install VS Code from the following link:
 - <https://code.visualstudio.com/download>



Install VS Code² (Windows)

- Open the .exe file and install VS Code
- **Important:** Accept all default settings.

Install VS Code² (MacOS)

- If archive, extract the archive contents (e.g., VSCode-darwin-universal.zip).
- Drag **Visual Studio Code.app** to the **Applications** folder

Install Anaconda Navigator

Anaconda

- Anaconda is a distribution of **Python** that includes tools and packages geared towards scientific computing (such as data science and machine learning)
- **Anaconda Navigator** is the graphical user interface (GUI) allowing users to install and manage their programming environment without command line (terminal) prompts



Install Anaconda

- You can install Anaconda from the following link:
 - <https://www.anaconda.com/download>

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Distribution

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For installation assistance, refer to [Troubleshooting](#).

Download Anaconda Distribution or [Miniconda](#) by choosing the proper installer for your machine. Learn the difference from our [Documentation](#).



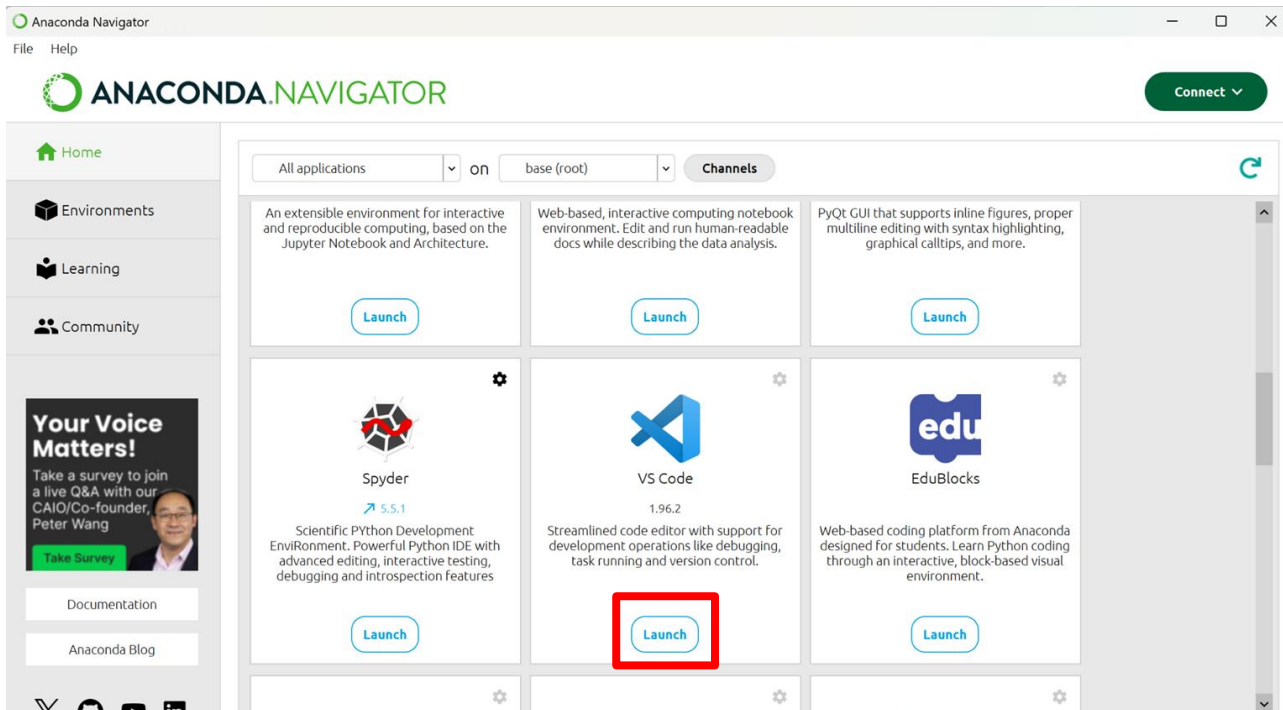
Anaconda Installers

 [Download](#)



Anaconda Navigator

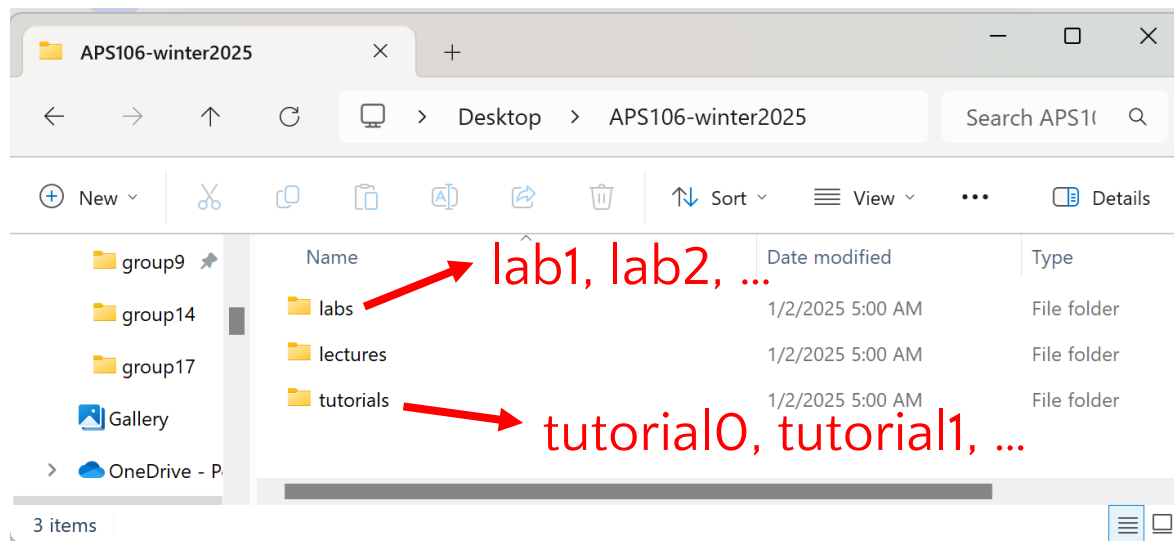
- We will be mainly using “**VS Code**” in this course. Let’s click on “Launch”!



Set up a Folder Structure for APS106

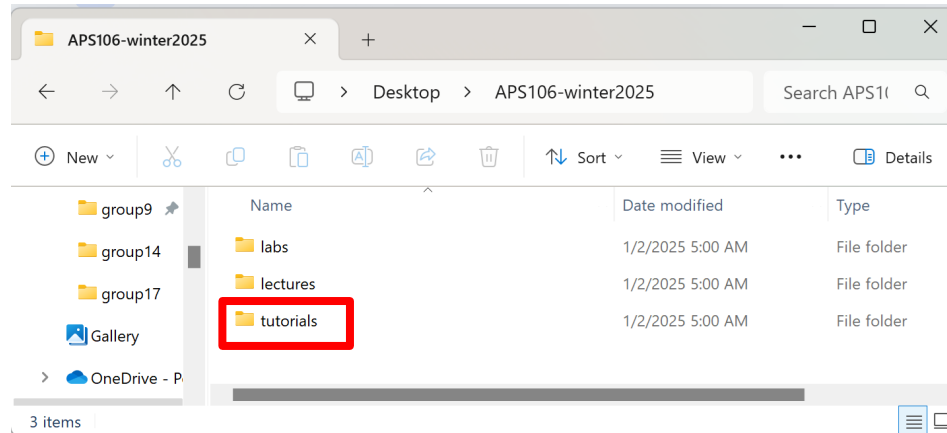
Stay Organized, Stay Efficient

- It's not a good idea to store all your files in the Downloads folder!
- Let's create a folder for APS106 (e.g., on Desktop) and organize your files there



Let's download files from Quercus!

- Go to Quercus → Modules → **Tutorial Homepage**
 - Click on **Tutorial 0 – Setting Up A Development Environment** in week1
 - Download “**tutorial0.zip**”
 - Unzip the zip file and move it in your folder for APS106



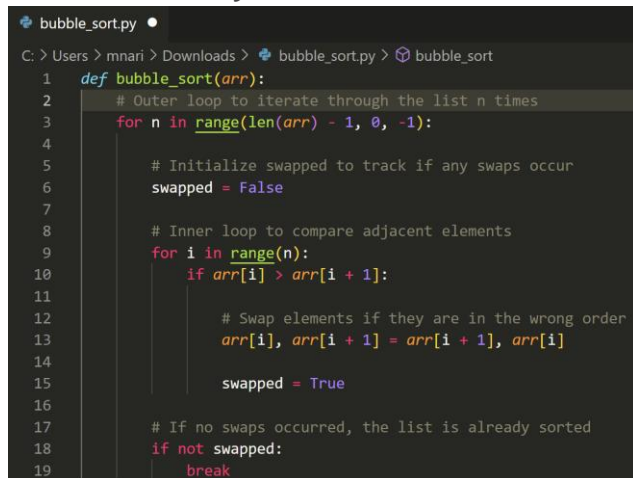
VSCode/Jupyter Notebook

VSCode is an IDE

- Integrated Development Environments (IDEs) are programs that provides tools and features to programmers in a unified environment. Think of them as "smart" text editors specifically designed for writing code
- IDEs often include:
 - **A code editor**
 - A place to type and edit code, usually with colour-coded syntax highlighting to improve readability
 - **Code compilers or interpreters**
 - Turns the readable Python code into something the machine can understand
 - **Debuggers**
 - Pause the code at pre-determined locations and go line-by-line through your code
- So IDEs basically contain **everything** you need to code!

VSCode is an IDE

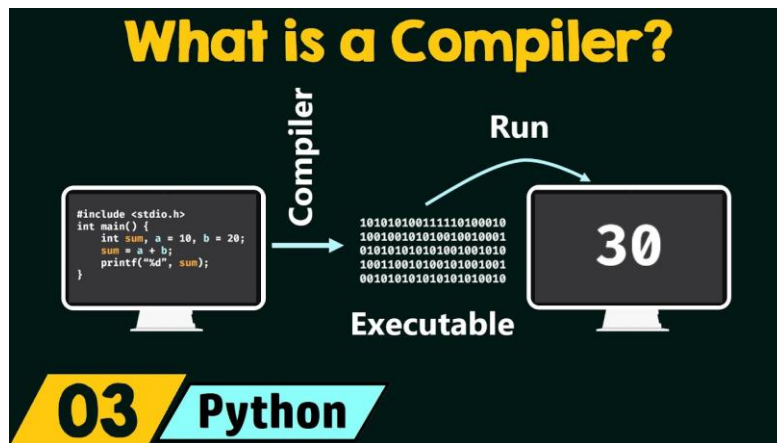
- Integrated Development Environments (IDEs) are programs that provides tools and features to programmers in a unified environment
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```
def bubble_sort(arr):  
    # Outer loop to iterate through the list n times  
    for n in range(len(arr) - 1, 0, -1):  
        # Initialize swapped to track if any swaps occur  
        swapped = False  
  
        # Inner loop to compare adjacent elements  
        for i in range(n):  
            if arr[i] > arr[i + 1]:  
                # Swap elements if they are in the wrong order  
                arr[i], arr[i + 1] = arr[i + 1], arr[i]  
  
                swapped = True  
  
        # If no swaps occurred, the list is already sorted  
        if not swapped:  
            break
```


VSCode is an IDE

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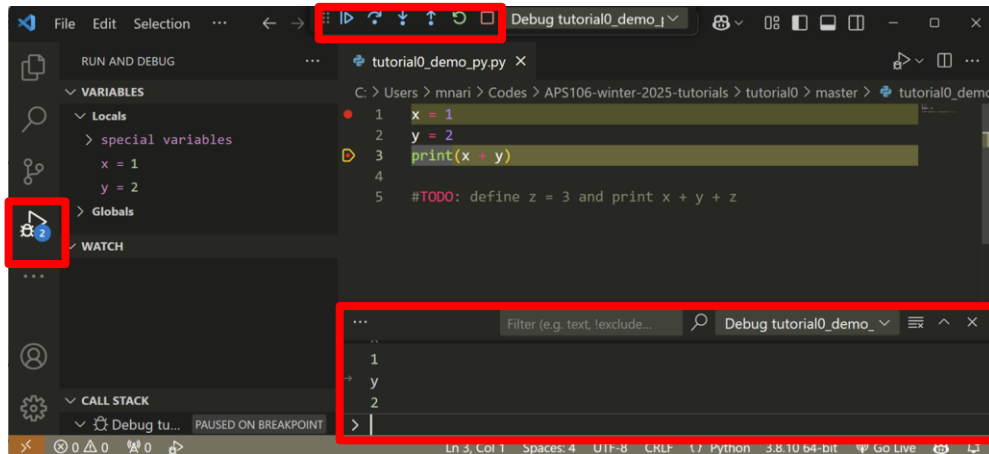


VSCode is an IDE

- Integrated Development Environments (IDEs) are programs that provides tools and features to programmers in a unified environment
- IDEs often include:
 - **Debuggers**
 - Pause the code at pre-determined locations and go line-by-line through your code

Continue, step over, step in/out, restart, stop

Run and Debug



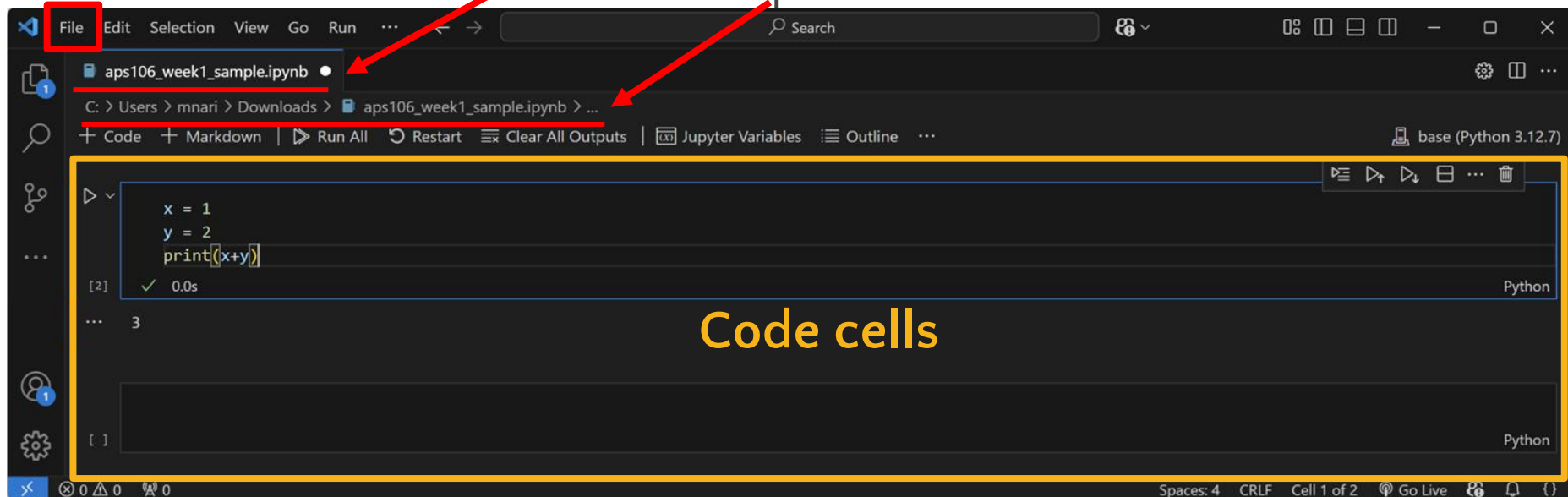
Debug console

VS Code (Jupyter Notebook)

Open/Save files

Filename

File path

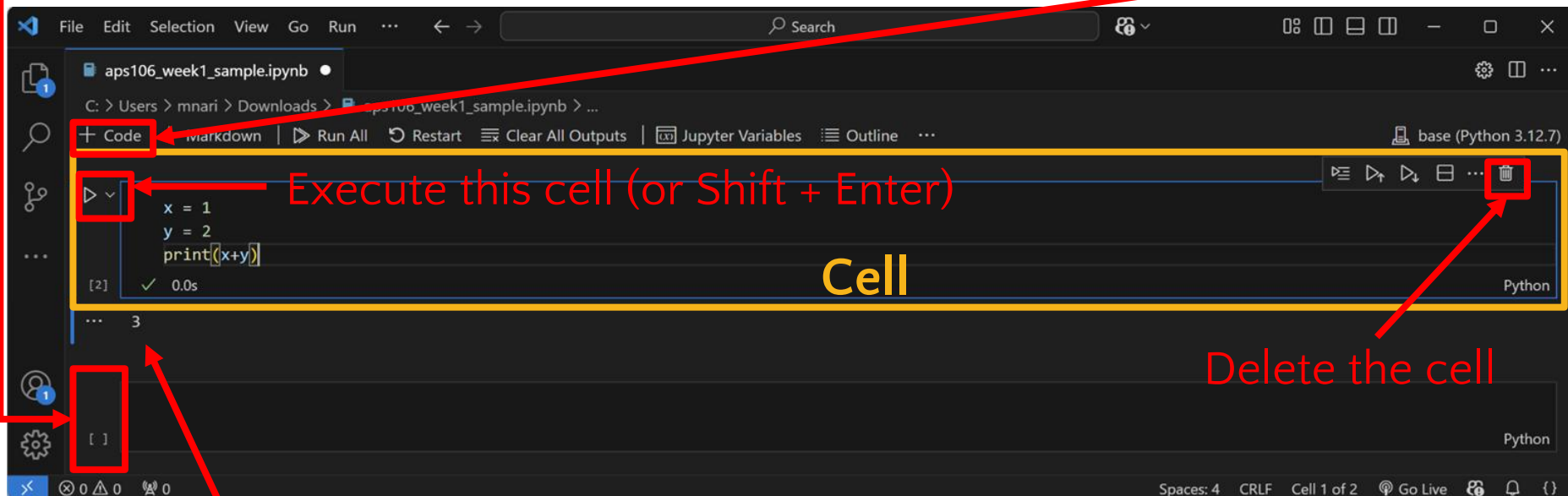


VS Code (Jupyter Notebook)

(drag&drop or Alt + ↑ / Alt + ↓)

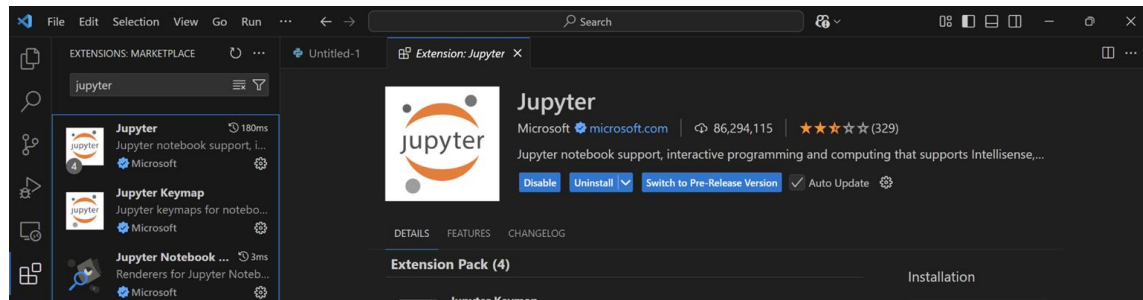
Move up/down the cell

Create a new cell under the current one



VS Code (Jupyter Notebook)

- Make sure you have “**Python**” and “**Jupyter**” extensions installed!



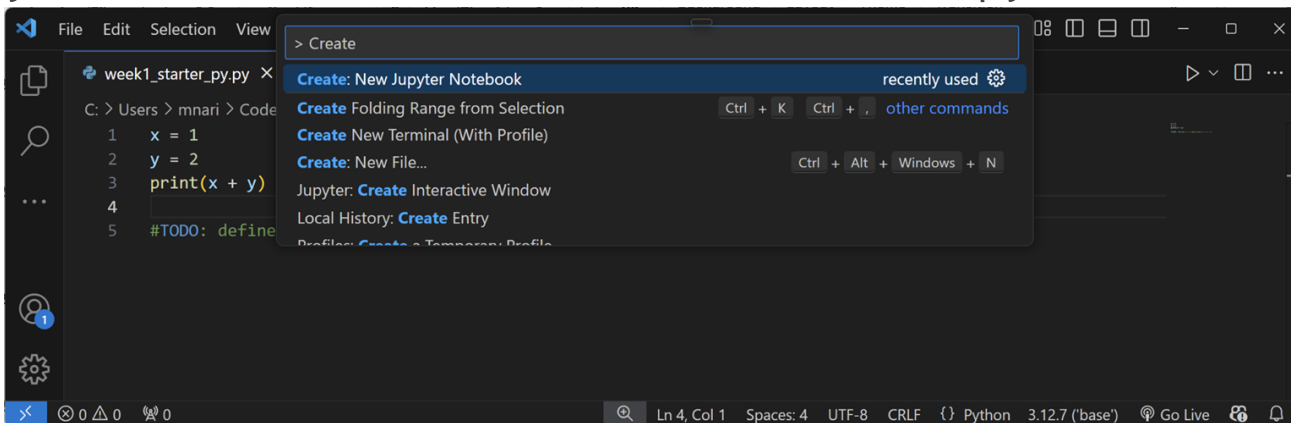
Let's practice!¹

- Let's open **tutorial0_demo_jupyter.ipynb**, write our first program, and save it!
- Steps:
 1. Launch VS Code through Anaconda Navigator
 2. File -> Open File -> Go to the APS106 folder and select **tutorial0_demo_jupyter.ipynb**
 3. In the first code cell, write

```
print("Hello World!")
```
 1. Execute the cell (if asked, select "base (Python 3.12.7)")
 2. File -> Save as -> save it as **"tutorial0_demo_jupyter.ipynb"**

Create a new Jupyter Notebook

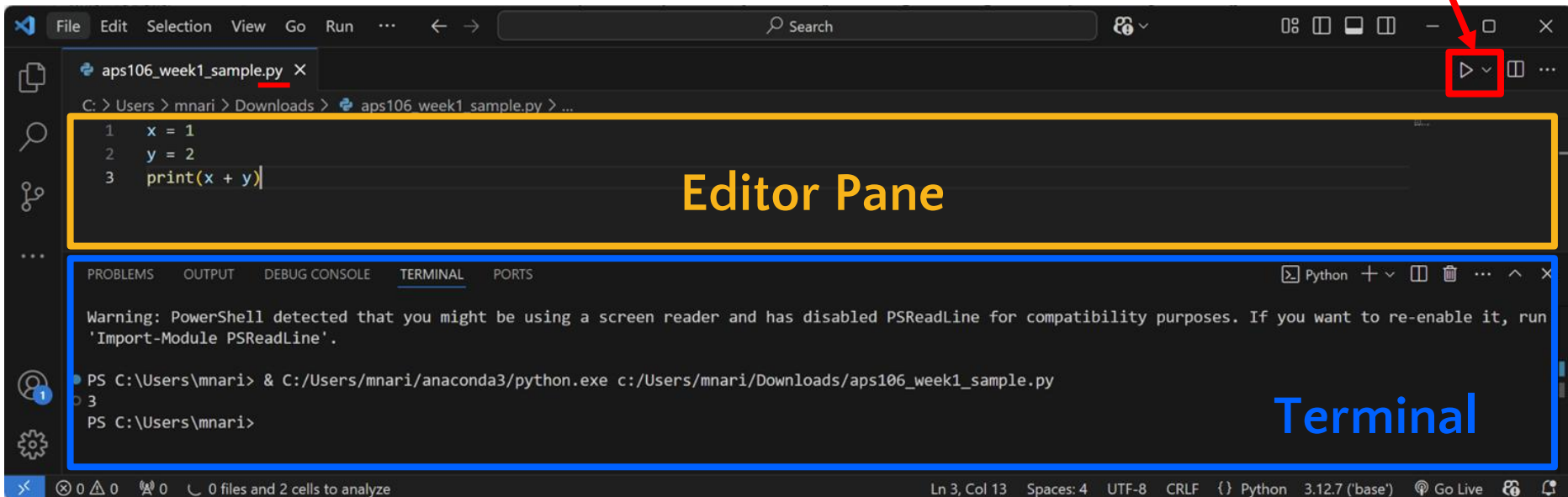
- You can create a new notebook on VS Code too!
- Steps:
 1. Launch VS Code through Anaconda Navigator
 2. In the command palette (“Search” on top), type in “> **create new jupyter notebook**” and select “Create: New Jupyter Notebook”



VS Code (Python script)

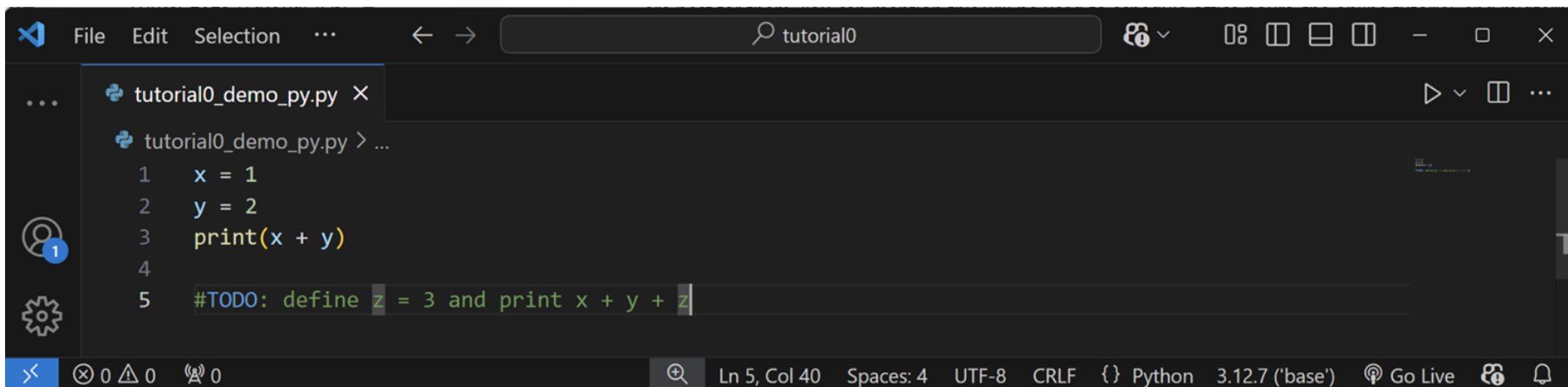
- When you open a `.py` file, an editor with a different layout appears

Execute the Python script



Let's practice!²

- Let's open **tutorial0_demo_py.py** in VS Code, modify the code based on the instruction, and run it!



```
File Edit Selection ... ← → tutorial0
```

```
... tutorial0_demo_py.py X
```

```
... tutorial0_demo_py.py > ...
```

```
1 x = 1
```

```
2 y = 2
```

```
3 print(x + y)
```

```
4
```

```
5 #TODO: define z = 3 and print x + y + z
```

```
< 0 0 0 Ln 5, Col 40 Spaces: 4 UTF-8 CRLF {} Python 3.12.7 ('base') Go Live
```

We'll switch from .ipynb to .py files later

- Jupyter Notebooks (.ipynb) are interactive and beginner-friendly
- Python scripts (.py) are **the standard format** for Python programs and are suited for building and running **larger projects**
- We'll switch to .py files in **lab5!**

APS106



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UofT JupyterHub

What is UofT JupyterHub?

- Access at <https://jupyter.utoronto.ca/>
- Cloud-based Jupyter Notebook service that allows us to run Jupyter Notebooks (.ipynb files) directly from a web browser
 - Don't need to install anything
- Linked to your UofT account, all lecture notes are stored as a copy in JupyterHub!

Let's try UofT JupyterHub!

- Go to **Tutorial Homepage** in APS106 Modules on Quercus
 - Click on **Tutorial 0 – Setting Up A Development Environment** in week1
 - Click on **JupyterHub Starter Link**

jupyterhub tutorial0_starter (autosaved) Python 3 (ipykernel) O

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel) O

Memory: 187.2 MB / 2 GB

Jupyter Notebooks are split into discrete "cells". This is a cell, and so is the block of text above, and the piece of Python code below.

Each cell can be run independently by hitting the Run button in the top toolbar, going to Cell -> Run Cells, or pressing "Shift + Enter"

Code cells contain executable Python code, as seen below.

TODO: Let's run the code block below. If you see "Hello World" printed below the cell, you can move on to the next problem!

```
In [ ]: print("Hello World!")
```